

Perfect Mode Hal: Aggressive, Change-Aware Opponent Exploitation in Exact DTH

We revised the August 20, 2026 note to add the later studies and the hosting decision. The PM v3 result is unchanged.

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Abstract

Perfect Mode (PM) Hal is a causal meta-policy for repeated, pure Drop the Handkerchief (DTH). One certified tablebase remains the sole authority for each continuation-adjusted 60×60 matrix. Around it, PM combines Adaptive Hal’s Bayesian counts, Perfect Hal v1’s pattern experts, online change-point and outcome models, and a compatible Aggro Hal recurrent network. Fixed Share learns role-specific source and controller weights. Four modes move from an exact shield to hard forecast responses, while every selected policy receives a fresh local worst-case-loss measurement. An earlier 24-identity development run scored 44 wins in 48 games, but its protocol lacked durable registration and its evaluator trusted self-reported risk. It has no promotion authority. This note instead registers a 56-identity confirmation protocol with independent matrix recomputation, a no-Aggro ablation, and family and seat gates. At the protocol commit that confirmation had not been executed. The one-shot run scored PM at .8705 (97–14–1), behind Perfect v1 at .9554; the promotion gate failed despite zero independent local-risk audit violations. PM is an aggressive candidate, not human mind reading, dominance, or a whole-game safety theorem. We registered a successor study of a one-step exploit continuation for translated Hal with a per-game guard on realized gifts; its selected arm failed three of six promotion gates. Under its decision rule we chose translated Hal v1, which a preview deployment serves, and we have not promoted that preview to production.

1 Scope and design philosophy

At public state x , the exact DTH agent supplies matrix $M(x)$, equilibrium policies (x^*, y^*) , and a saddle-gap certificate at most 10^{-6} . All actions are literal seconds $1, \dots, 60$. PM never estimates continuation values and never sees an unrevealed simultaneous action.

The intended character is “overbearing after evidence, composed after surprise.” Perfect Hal v1, which later records call Old, contributes hard exploitation; Adaptive contributes uncertainty and a constrained response frontier; Aggro contributes recurrent features and a learned residual policy. The story evidence ledger motivates competing personae, opponent mental models, and abrupt memory discontinuities. PM translates that fiction into testable

engineering—model competition, role separation, recency, and change recovery—rather than treating a person as one fixed psychological type. The final narrative ideal, to improve and adapt without looking back, becomes persistent memory with rapid de-weighting and recovery after changed play.

2 Causal opponent model

For each opponent role separately, PM maintains six mandatory forecast sources: uniform recovery, the opponent’s exact equilibrium policy, a decayed Dirichlet model, Perfect v1’s online expert mixture, categorical Bayesian online change-point detection (BOCPD), and an outcome-conditioned model. A seventh source is Aggro’s GRU forecast when a strict compatible checkpoint is present. After action a_t is revealed, source k receives prequential log score and Fixed Share updates

$$\begin{aligned}\tilde{w}_{k,t+1} &\propto w_{k,t} \exp(\eta \log q_{k,t}(a_t)), \\ w_{k,t+1} &= (1 - \alpha) \tilde{w}_{k,t+1} + \frac{\alpha}{K - 1} (1 - \tilde{w}_{k,t+1}),\end{aligned}$$

with $\alpha = .04$. Thus every source retains recovery mass when an opponent switches. BOCPD carries a truncated Dirichlet–multinomial run-length posterior with hazard .08. The fused forecast is $\hat{q}_t = \sum_k w_{k,t} q_{k,t}$.

Aggression confidence is the declared heuristic

$$c_t = c_{\max} \frac{n_{\text{eff}}}{4 + n_{\text{eff}}} (1 - P_t(\text{change})) (1 - \text{JS}_t) s_t,$$

where JS_t is normalized source disagreement and $s_t = \exp[-\max(0, \overline{\text{NLL}}_t - \log 60)]$. Only the evidence fraction has a data-biased-response interpretation; the product is a project controller, not a published theorem.

3 Aggression with measured risk

For Dropper, the certified ϵ candidate solves

$$\max_{x \in \Delta_{60}} x^\top M \hat{q}_t \quad \text{s.t.} \quad (M^\top x)_j \geq L - \epsilon \quad \forall j,$$

where $L = \min_j (M^\top x^*)_j$ is freshly recomputed. Checker uses the symmetric upper constraint. The bank also contains exact equilibrium, hard responses to fused and Perfect

v1 forecasts, and Aggro’s direct policy. For a direct Dropper policy x , actual local loss is $\max(0, L - \min_j (M^\top x)_j)$; every direct and frontier candidate is independently checked against the current matrix.

One common bank of heuristic ensemble perturbations must give positive expected improvement and at least .55 positive-draw support. These draws are not a calibrated posterior, so the support fraction is not an improvement probability. Cold start or a change shock enters *shield*, where only the exact face is admissible. The registered v3 caps are *probe* .05, *press* .50, and *dominate* 2.0, inside a 12.0 per-game charge budget. Controller weights receive full-information counterfactual payoff feedback after reveal. This is deliberately aggressive: the budget is an explicit risk appetite, not a guarantee that losses add across a DTH game. Ganzfried–Sandholm’s cumulative ledger concerns repetitions of one stage game; DTH’s matrices change with the state and already contain overlapping continuation values.

4 Development audit and registered confirmation

The v1 and v2 files were run before this branch registered them in Git; mutable status text cannot establish preregistration. They remain development evidence. V2 used 24 identities across six families, common seeded seat pairs, and 10,000 identity-clustered bootstrap replicates. Stopped games scored .5.

Table 1: Development v2 and registered v3 outcomes.

Controller	V2 score	V3 score	V3 W–L–S
PM Hal	.9167	.8705	97–14–1
Perfect v1	.8958	.9554	107–5–0
PM no Aggro	—	.8259	92–19–1
Aggro	.6458	.7232	81–31–0
Adaptive	.6563	.7188	80–31–1
Exact	.5417	.4955	54–55–3

V2’s nominal every-component gate failed. Its controller-reported risk was not independently reconstructed, so it cannot promote PM. Fused expected NLL was 3.481 [2.904, 3.978], but change detection found only 10 of 86 truth shifts. These mixed results motivated the registered repair.

Clean-Git v3 hash-binds the controller, recurrent checkpoint, and tablebase, records its commit, and uses 56 fresh identities over all 14 families (112 games per controller). It compares PM with Exact, Perfect v1, Adaptive, Aggro, and PM without Aggro. Promotion requires zero independent risk violations and nonnegative aggregate, both-seat, and every-family comparisons. Every selected policy’s loss, charge contract, mode cap, cumulative charge, and game budget is recomputed from its continuation-adjusted matrix.

The registered protocol ran once from commit `b76147f`.

PM-minus-baseline paired score differences were $-.0848$ $[-.1518, -.0223]$ against Perfect v1, $+.0446$ $[.0089, .0893]$ against PM without Aggro, and $+.3750$ $[.2723, .4777]$ against Exact. The gate failed the every-component, both-seat, and every-family conditions; Perfect v1 was the strongest component. The independent audit found zero violations over 788 decisions and 112 games, with maximum measured local loss 1.5373. Fused expected NLL was 3.5524 versus uniform 4.0943, but the change detector found only 35 of 168 truth shifts and produced 23 false-alarm onsets. The result supports aggregate value from enabling the Aggro component in this protocol; it does not isolate recurrent memory as the cause. The hash-bound compact record is `src/arena/config/pm_hal_confirmation_v3.json`, and the evaluator is `src/hal_lab/experiments/pm_hal_v3/evaluate_pm_hal.py`.

We rebuilt the tablebase after v3, and the later builds replaced the manifest `src/dth/artifacts/complete_full_v1/tablebase.json` that v3 binds by SHA-256. The provenance check `python -m hal_lab.provenance --check` reports that binding as a recorded mismatch. The v3 record sets `v3_must_not_be_rerun_or_rebound`, so we do not rerun or rebind v3. A new question requires a new protocol version.

PM runs in local pure-DTH play, and the hosted game does not offer it. The hosted runtime loads no torch, and `STL_HAL_POLICY` accepts `exact` and two exploit policies, `translated-v1` and `exploit-continuation-v1`.

5 Successors to Perfect v1

Two development studies built on Perfect v1. Translated Hal v1 adds offset experts to Old’s expert mixture. Each offset expert copies or mirrors one of Hal’s own revealed actions. The selector study trains an attention network that corrects translated Hal’s 24 expert weights. Table 2 gives their headline paired comparisons from `translated_hal_v1_results.json` and `selector_study_v1_results.json` in `src/hal_lab/experiments/`. The emulator comparison uses eight browser identities, and its interval includes zero.

Table 2: Development evidence after Perfect v1. Scores are win fractions. Diff. gives paired differences with identity-clustered 95% intervals.

Comparison	Games	Cand.	Base	Diff.	[95%]
Transl. vs. Old, league	8,192	.915	.894	+.021	[.016, .027]
Transl. vs. Old, emulators	5,120	.903	.902	+.001	[−.003, .006]
Attention vs. transl.	2,432	.919	.904	+.015	[.005, .026]

These leagues and emulators differ from the v3 league, so no number in Table 2 transfers to Table 1. The translated record’s `worktree` field holds a disclaimer of Git preregistration. Git first holds both records and their runners in commit `4944c91` of September 24, 2026, after the runs. Neither study has Git preregistration, and we treat both as development evidence.

6 One-step exploit continuation

Method. Translated Hal answers its forecast q in the exact stage matrix, and that matrix values each child state at equilibrium. The provider `src/arena/policies/exploit_continuation.py` replaces each live child’s equilibrium value with the value of Hal’s best response to a child forecast q' at that child. The *same* forecast reuses q . The *role* forecast asks the opponent model for the role that the opponent takes at the child. Hal then plays a hard best response to q in the look-ahead matrix L .

Let H be the Hal-view exact matrix: M for Dropper, $-M^\top$ for Checker. At a child, Hal can guarantee the maximin value of the child’s stored matrix against every opponent policy, so Hal’s best response to the one policy q' earns at least that value. By the certificate, that maximin value lies at most 10^{-6} below the stored child value that H uses. Each entry of L weights child values with the same revival odds as H , and terminal children keep their values. Thus $L \geq H$ entry by entry, up to the 10^{-6} certificate tolerance.

Guard. The matrix L assumes that the opponent plays q' at every child. The guard limits the cost of that assumption in each game. Let $W_H(s)$ be the stored Hal-view value, with $W_H = \pm 1$ at a terminal win or loss, and let $H_t[a, b] = \mathbb{E}[W_H(s_{t+1}) \mid s_t, a, b]$. Hal carries a budget k_t with $k_0 = \epsilon_0$ and the floor $f_t = W_H(s_t) - \max(k_t, 0)$. If the best response π_t gives $\min_b(\pi_t^\top H_t)_b < f_t$, a HiGHS LP maximizes $\pi^\top Lq$ subject to $\pi^\top H_t \geq f_t$. If the LP fails, Hal plays its stored equilibrium policy. After the reveal of b_t , the budget absorbs the realized gift:

$$g_t = \pi_t^\top H_t e_{b_t} - W_H(s_t), \quad k_{t+1} = k_t + g_t.$$

The certificate lets the maximin value of H_t lie up to 10^{-6} below $W_H(s_t)$, so at some states no policy reaches f_t . The turn’s miss $m_t = \max(0, f_t - \min_b(\pi_t^\top H_t)_b)$ adds to the game’s slack $S = \sum_t m_t$.

Bound. In pure DTH, against any opponent, a guarded Hal has $\sum_t g_t \geq -\epsilon_0 - S$ on every path and $\mathbb{E}[G] \geq W_H(s_0) - \epsilon_0 - \mathbb{E}[S]$, where $G = \pm 1$ is the outcome.

Proof. Each live DTH transition raises a bounded potential Φ (`src/formal/CLAIMS.md`), so each path ends at a terminal state s_T with $W_H(s_T) = G$. The outcome telescopes:

$$G - W_H(s_0) = \sum_{t < T} (W_H(s_{t+1}) - W_H(s_t)).$$

The opponent commits b_t without sight of a_t , and Hal draws a_t from π_t . Given the history and b_t , term t thus has the conditional mean $\pi_t^\top H_t e_{b_t} - W_H(s_t) = g_t$. The floor gives $g_t \geq -\max(k_t, 0) - m_t$, so $k_{t+1} \geq \min(k_t, 0) - m_t$. Induction from $k_0 = \epsilon_0 \geq 0$ gives $k_t \geq -\sum_{u < t} m_u$. At the end, $\sum_t g_t = k_T - \epsilon_0 \geq -\epsilon_0 - S$. Each term minus g_t has conditional mean zero, and the horizon is bounded, so $\mathbb{E}[G] - W_H(s_0) = \mathbb{E}[\sum_t g_t] \geq -\epsilon_0 - \mathbb{E}[S]$. $\square$

In confirmation, the largest game slack was 9.8×10^{-10} . The score is $(1 + G)/2$, and the two seats’ stored scores at clock 720 are .5449 and .4551. A paired-seat session thus

has the bound $.5 - \epsilon_0/2$. The Counter column of Table 3 shows it for each guard. The caveat of Section 3 concerns PM’s sum of local worst-case charges, whose continuation values overlap. That caveat does not cover this budget, because realized gifts telescope through the stored values. Canonical leap turns use the equilibrium fallback and leave k unchanged, so the bound covers pure DTH.

Primary metric. The study harness reads each simulated opponent’s mixed policy μ_t . It scores a finished pure-DTH game by

$$\hat{s} = \frac{1}{2} \left(1 + W_H(s_0) + \sum_t (\pi_t^\top H_t \mu_t - W_H(s_t)) \right)$$

(`src/hal_lab/harness/telescope.py`). The same telescoping argument gives $\hat{s}$ the mean of the raw score. It removes the per-step noise of both players’ draws and of revival chance, and it can leave $[0, 1]$. Canonical games report the raw score, because leap turns change the transition.

Registered protocol. We pushed the runner `run_exploit_continuation.py` with its frozen PROTOCOL in commit `b8f87b4` before any run. The `record` phase checked that a remote branch holds those runner bytes. Commit `b218219` holds the results record `exploit_continuation_v1_results.json`. Both files live in `src/hal_lab/experiments/exploit_continuation_v1/`. Eight arms cross the two child forecasts with four guard settings: open, and $\epsilon_0 = .10, .25$, and $.50$. Pure-DTH sessions play four games from clock 720 in paired seats, with a cap of 240 half-rounds.

Pure-DTH play uses four opponent blocks. The league holds 19 families. The league mean covers the 18 non-equilibrium families, and we report the equilibrium family apart. The human block fits three emulator styles to the train games of each of 8 players. The Hal-model block plays the selected translated Hal as the opponent. The counter block runs a private copy of the controller on the public history and best-responds to its exact mixed policy. Canonical sessions play eight STL games against the emulators, with the candidate in the Hal seat. Bootstraps use 10,000 replicates, clustered by player in human blocks and by identity elsewhere.

The canonical phase (seed base 118,300,000) compared exact and translated Hal with 64 emulator replicates. Selection (118,600,000) played every arm beside both baselines, with 24 league identities per family and 8 emulator replicates. Each Hal block in selection had 64 identities. Confirmation (119,100,000) doubled the league and Hal blocks and used 24 emulator replicates. It also added 16 canonical replicates. To select, we keep the arms whose league mean $\hat{s}$ reaches translated Hal’s and whose Hal-model mean $\hat{s}$ is at most .025 below translated Hal’s. From those, we pick the highest human mean $\hat{s}$. Table 4 lists gates P1 to P6. If every gate passes, we deploy the arm. If a gate fails, we follow the canonical gate: we deploy translated Hal when its canonical human score beats exact Hal’s with a player-clustered 95% lower bound above 0, and exact Hal otherwise.

Table 3: Selection mean $\hat{s}$ by block. Rule floors: .9605 (league) and .4736 (Hal model). One arm passed: role, $\epsilon_0 = .25$.

Controller	League	Human	Hal model	Counter
Exact	.500	.500	.500	.500
Translated	.961	.910	.499	-.007
Same, open	.993	.948	.339	-.025
Same, $\epsilon_0=.10$	.906	.819	.511	.450
Same, $\epsilon_0=.25$	.984	.907	.471	.375
Same, $\epsilon_0=.50$	.992	.934	.426	.253
Role, open	.993	.940	.401	-.018
Role, $\epsilon_0=.10$	.904	.815	.517	.450
Role, $\epsilon_0=.25$	.984	.905	.489	.376
Role, $\epsilon_0=.50$	.992	.930	.440	.255

Table 4: Confirmation gates: role, $\epsilon_0 = .25$, minus translated Hal. P1–P3 and P5 pass when the 95% lower bound exceeds the limit.

Gate	Block, metric	Mean	Lower	Limit	Result
P1	Human, $\hat{s}$	+.0001	-.0166	0	fail
P2	League, $\hat{s}$	+.0346	+.0260	-.005	pass
P3	Hal model, $\hat{s}$	-.0044	-.0293	-.025	fail
P5	Canonical, raw	-.0029	-.0218	-.010	fail
P4	Guard, $\sum_t g_t$	—	—	—	pass
P6	Runtime, p95	—	—	—	pass

P4: $\min \sum_t g_t = -.2500000001$ over 6,976 games, with a largest turn miss of $9.8 \times 10^{-10} \leq 10^{-6}$. P6: p95 decision and request times of 24.9 and 25.0 ms (limits 100 and 300); p95 worker recovery 243.3 ms (limit 2,000).

Results. Translated Hal scored .9394 (11,543–745) and exact Hal .5467 (6,718–5,570) over 12,288 canonical games each. The player-clustered difference was +.3927 [.3704, .4073], so the canonical gate passed.

Table 3 gives the selection. In confirmation, the chosen arm scored .990 on the league against translated Hal’s .955. It tied translated Hal on the emulators at .911, and it scored .486 against .490 on the Hal model. It failed P1 and P3 on pure-DTH blocks and P5 in canonical play (Table 4). The runtime check `src/browser/deploy/check_exploit_continuation.py` passed P6 in a local run of the hosted app with the real tablebase. That run excludes Redis latency and cold starts.

Against its full-knowledge counter, translated Hal lost every game: 0–256 in selection and 0–512 in confirmation. The counter reads the public history and runs the provider source, so any player who runs the public code can predict and answer each mixed policy of translated Hal. Against its own counter, exact Hal scored $\hat{s} = .500$ (raw .465, 238–274).

Interpretation. The guarded arms reached the bound $.5 - \epsilon_0/2$ against the counter (Table 3). The open arms gained on the league and the human emulators and fell against the Hal-modelling opponent, with Hal-model scores of .339 and .401 against translated Hal’s .499. The $\epsilon_0 = .10$ arms held against the Hal model and lost on the league and the human emulators. No guard setting gained on the human emulators and held against the Hal model. Under the decision rule, we chose **translated-v1** for deployment.

7 Deployment status and human-play limits

As of September 29, 2026, a preview deployment serves translated Hal v1, and we have not promoted that preview to production.

The psychology is operational: reinforcement, belief learning, response to one’s own revealed action, recency, periodicity, and change are competing causal hypotheses. A human can counter-adapt, deceive, or change strategy; PM must preserve recovery mass and retreat on surprise. It should never label a participant’s personality or claim privileged access to intent.

A human study must keep opponent identity across games, log the sequence forecast before each reveal, score realized NLL/Brier without simulator-truth distributions, and cluster outcomes by participant. Change delay and false alarms, role slices, calibration, and game score are co-primary diagnostics. Bounded recursive reasoning is one possible population archetype, not a label the controller may assign permanently to a participant.

Every emulator result in this note comes from 8 players with nonempty test partitions. Each emulator fits one player’s train games, and those games all faced exact Hal. The data thus hold no human reaction to exploitation. We need a live randomized comparison, clustered by participant, before we claim a human win rate for any policy in this note.

The measured evidence comprises the PM development run over 24 identities, the registered PM v3 confirmation over 56 fresh synthetic identities, the development studies of Section 5, and the registered exploit-continuation study of Section 6. The recurrent checkpoint saw 2,653 training decisions. Its enabled component improved aggregate score over the no-Aggro ablation, while a causal recurrent-memory advantage remains open. PM is feature-complete for human play as an aggressive candidate. Synthetic promotion failed; any future protocol requires a new version and fresh identities, followed by participant-level human evidence.

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